UNITED ARAB EMIRATES
MINISTRY OF EDUCATION

الإمارات العربية المتحدة
وزارة التربية والتعليم

PHYSICS – IMPORTANT FORMULAE & CONSTANTS

GRADE 10 ADVANCED | TERM 1 | 2026–2027

MODULE 16 | REFLECTION & SPHERICAL MIRRORS

LAW OF REFLECTION

$$\theta_i = \theta_r$$

θ_i = incidence angle; θ_r = reflection angle. Measure both from the normal (degrees).

PLANE-MIRROR IMAGE

$$X_i = -X_o \quad h_i = h_o$$

X_o = object distance; X_i = signed image distance (m). Image is virtual, upright and same size.

MIRROR EQUATION

$$\frac{1}{f} = \frac{1}{X_o} + \frac{1}{X_i}$$

f = focal length (m); X_o, X_i = object and image distances (m).

FIND THE IMAGE DISTANCE

$$X_i = \frac{fX_o}{X_o - f}$$

Rearranged mirror equation. Use signed focal length and a consistent length unit.

MAGNIFICATION

$$m = \frac{h_i}{h_o} = -\frac{X_i}{X_o}$$

m = magnification (unitless); h_i, h_o = signed image and object heights (m).

FOCAL LENGTH & CURVATURE

$$f = \frac{R}{2}$$

R = signed radius of curvature (m). For a spherical mirror, the focus is halfway to the centre.

MODULE 16 | MIRROR SIGN CONVENTION

QUANTITY	POSITIVE	NEGATIVE
f, R	Concave mirror	Convex mirror
X_i	Real image: in front	Virtual image: behind
h_i, m	Upright image	Inverted image

Size: |m| > 1 enlarged; |m| = 1 same size; |m| < 1 reduced. A real object in front has $X_o > 0$. A convex mirror always gives a virtual, upright, reduced image.

MODULE 16 | REFRACTION, SNELL'S LAW & CRITICAL ANGLE

REFRACTIVE INDEX

$$n = \frac{c}{v}$$

n = refractive index (unitless); c = light speed in vacuum; v = speed in the medium (m/s).

SPEED IN A MEDIUM

$$v = \frac{c}{n}$$

c = 3.00 × 10^8 m/s. Higher n means lower light speed.

SNELL'S LAW

$$n_1 \sin\theta_1 = n_2 \sin\theta_2$$

θ_1, θ_2 = angles to the normal; n_1, n_2 = refractive indices.

FIND THE REFRACTION ANGLE

$$\theta_2 = \sin^{-1}\left(\frac{n_1 \sin\theta_1}{n_2}\right)$$

Use degree mode. If the sine value would exceed 1, no refracted ray is possible.

CRITICAL ANGLE

$$\sin\theta_c = \frac{n_2}{n_1}$$

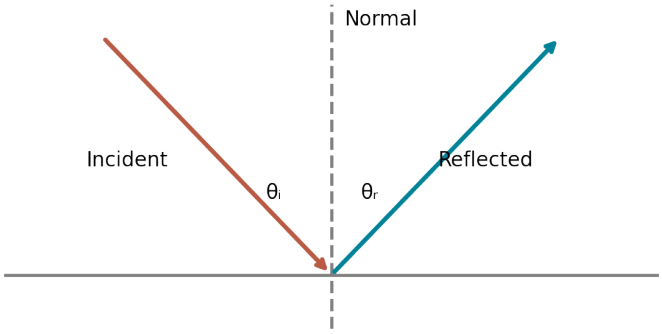
Only for $n_1 > n_2$. At θ_c, the refracted ray is at 90° to the normal.

WAVELENGTH IN A MEDIUM

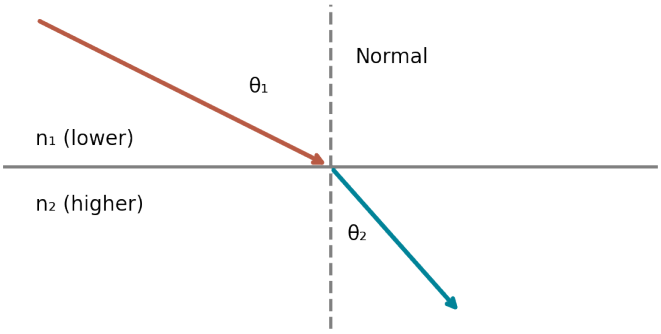
$$\lambda_{\text{medium}} = \frac{\lambda_0}{n}$$

λ_0 = vacuum wavelength (m). Frequency does not change when light crosses a boundary.

Reflection: angles from the normal



Refraction: toward normal if n increases



QUICK RULE	MEANING
Total internal reflection	$n_1 > n_2$ and $\theta_1 > \theta_c$. At $\theta_1 = \theta_c$, light travels along the boundary.
Bending / dispersion	Into higher n: toward normal; into lower n: away from normal. In glass, violet usually bends more than red.

MODULE 16 | THIN LENSES — IMAGE POSITION & HEIGHT

THIN-LENS EQUATION

$$\frac{1}{f} = \frac{1}{X_o} + \frac{1}{X_i}$$

f = lens focal length (m); X_o = object distance (m); X_i = signed image distance (m).

IMAGE DISTANCE

$$X_i = \frac{fX_o}{X_o - f}$$

For a real object, $X_o > 0$. Use the sign of f appropriate to the lens.

OBJECT DISTANCE

$$X_o = \frac{fX_i}{X_i - f}$$

This rearrangement also applies to the mirror equation.

FOCAL LENGTH

$$f = \frac{X_o X_i}{X_o + X_i}$$

Use the same unit for all lengths. Do not replace signed distances with their magnitudes.

LENS MAGNIFICATION

$$m = \frac{h_i}{h_o} = -\frac{X_i}{X_o}$$

m > 0: upright; m < 0: inverted. |m| gives the size ratio.

IMAGE HEIGHT

$$h_i = mh_o = -\frac{X_i}{X_o}h_o$$

h_i = signed image height; h_o = object height (same length unit).

MODULE 16 | LENS IMAGE GUIDE

LENS / OBJECT POSITION	IMAGE
Convex: $X_o > 2f$	Real, inverted, reduced; $f < X_i < 2f$.
Convex: $X_o = 2f$	Real, inverted, same size; $X_i = 2f$.
Convex: $f < X_o < 2f$	Real, inverted, enlarged; $X_i > 2f$.
Convex: $X_o = f$	Emergent rays parallel; no finite image position.
Convex: $0 < X_o < f$	Virtual, upright, enlarged; $X_i < 0$.
Concave lens: any real object	Virtual, upright, reduced; $f < 0$ and $X_i < 0$.

Signs: convex (converging) lens: f > 0; concave (diverging) lens: f < 0. $X_i > 0$ on the opposite side from the object (real); $X_i < 0$ on the object side (virtual). **Vision:** myopia → diverging lens; hyperopia → converging lens.

MODULE 17 | DOUBLE-SLIT INTERFERENCE

BRIGHT-FRINGE CONDITION

$$d\sin\theta = m\lambda$$

d = slit separation (m); λ = wavelength (m); m = 0, 1, 2, …; θ = angle from the central axis.

BRIGHT-FRINGE POSITION

$$x_m \simeq \frac{m\lambda L}{d}$$

x_m = distance from central bright band (m); L = screen distance (m). Small-angle approximation.

FIRST BRIGHT FRINGE / WAVELENGTH

$$\lambda \simeq \frac{x_1 d}{L}$$

x_1 = central-to-first-bright distance (m). For order m: $\lambda \approx x_m d/(mL)$.

FRINGE SPACING

$$\Delta x \simeq \frac{\lambda L}{d}$$

Δx = adjacent bright-fringe spacing (m). Larger λ or L widens the pattern.

DESTRUCTIVE INTERFERENCE

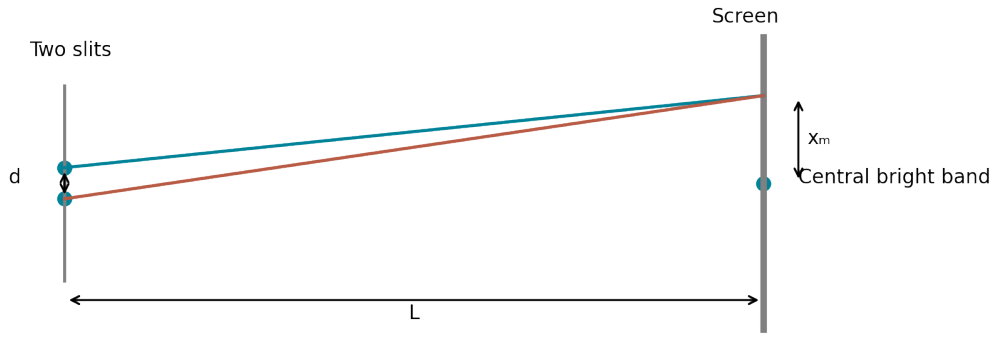
$$\Delta\ell = (m + \frac{1}{2})\lambda$$

Δl = path difference (m); m = 0, 1, 2, … . Waves arrive half a cycle out of step.

CONSTRUCTIVE INTERFERENCE

$$\Delta\ell = m\lambda$$

Δl = path difference (m); waves arrive in phase. Central maximum has m = 0.



MODULE 17 | THIN-FILM INTERFERENCE — REFLECTED LIGHT

PHASE INVERSIONS	BRIGHT (CONSTRUCTIVE)	DARK (DESTRUCTIVE)
Exactly one	$2nt = (m + ½)\lambda_0$	$2nt = m\lambda_0$
Both or neither	$2nt = m\lambda_0$	$2nt = (m + ½)\lambda_0$

Normal incidence: t = film thickness (m); n = film index; λ_0 = vacuum wavelength (m); m = 0, 1, 2, … (choose a positive physical thickness). Lower-to-higher n reflection adds a 180° phase shift; higher-to-lower n adds none.

MODULE 17 | SINGLE-SLIT DIFFRACTION & GRATINGS

SINGLE-SLIT CENTRAL WIDTH

$$W = 2x_1 \simeq \frac{2\lambda L}{w}$$

W = full central bright-band width (m); w = slit width (m); L = screen distance (m). Small angles.

CENTRE TO FIRST MINIMUM

$$x_1 \simeq \frac{\lambda L}{w}$$

x_1 is half the central width. Do not confuse it with the full width W.

FIND WAVELENGTH / SLIT WIDTH

$$\lambda \simeq \frac{Ww}{2L} \qquad w \simeq \frac{2\lambda L}{W}$$

All lengths must use compatible units. Narrower slit → wider central band.

DIFFRACTION-GRATING MAXIMA

$$d\sin\theta_m = m\lambda$$

d = grating spacing (m); θ_m = order-m angle; m = 0, 1, 2, … . Do not assume small angles.

FIRST-ORDER GRATING LINE

$$\lambda = d\sin\theta_1$$

For m = 1 only. θ_1 is measured from the central maximum, not from another order.

GRATING SPACING

$$d = \frac{1}{N}$$

N = number of lines per metre (m^{-1}); d = spacing (m). Convert lines/mm to lines/m first.

QUICK REFERENCE | CONSTANTS, SYMBOLS & UNIT CONVERSIONS

SYMBOL / CONVERSION	MEANING / USE
c	3.00×10^8 m/s (vacuum light speed); air n ≈ 1.00.
λ, d, w, t, L, X_o, X_i, f	Lengths: SI unit m. Keep the signed values for image calculations.
n, m	Refractive index and magnification / order are unitless; distinguish m by context.
Optical lengths	1 nm = 10^{-9} m; 1 µm = 10^{-6} m; 1 mm = 10^{-3} m; 1 cm = 10^{-2} m.
Grating line density	N (lines/m) = 1000 × N (lines/mm). Example: 500 lines/mm → d = 2.00×10^{-6} m.
Before substituting	Angles in degrees; all lengths in compatible units; λ in vacuum unless stated otherwise.